

Computer Architecture  
Lab 3

You are required to build the following:

- A register file which contains 8 registers, each register has 8-bits widths. The register file has 2-read address and 1- write address. It also has 2- read ports and 1-write port. It also has a write enable and a reset signal. (What is the size of address bus? What is the size of data bus?)
- Use the given registers (DFF) to create the register files.
- Create testbench to do the following:
  - Reset all registers, set all read addresses to zero
  - Write in Reg(0) 0xFF
  - Write in Reg(1) 0x11
  - Write in Reg(7) 0x90
  - Write in Reg(3) 0x08
  - Read Reg(1) on port 0 , Reg(7) on port 1 and **write 0x03 Reg(4)**
  - Read Reg(2) on port 0 , Reg(3) on port 1
  - Read Reg(4) on port 0 , Reg(5) on port 1
  - Read Reg(6) on port 0 , Reg(0) on port 1 and **write 0x01 Reg(0)**

**Assignment:**

- Create the same register file using memory arrays instead of DFFs.
- Test your new register file using the same testbench.
- Use Quartus to generate RTL view for both designs to compare between them.

| CYCLE# | Read Port 0 | Read Port 1 |
|--------|-------------|-------------|
| 1      | 0           | 0           |
| 2      | FF          | FF          |

|          |           |           |
|----------|-----------|-----------|
| <b>3</b> | <b>FF</b> | <b>FF</b> |
| <b>4</b> | <b>FF</b> | <b>FF</b> |
| <b>5</b> | <b>FF</b> | <b>FF</b> |
| <b>6</b> | <b>11</b> | <b>90</b> |
| <b>7</b> | <b>00</b> | <b>08</b> |
| <b>8</b> | <b>03</b> | <b>00</b> |
| <b>9</b> | <b>00</b> | <b>01</b> |